

What-is-driving-low-employee-engagement-.R

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```
# Load the packages
library("readr")
library("dplyr")

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library("ggplot2")
library("tidyr")

# Import the data
survey <- read_csv("~/Desktop/Data/survey_data.csv" )

## Rows: 1470 Columns: 5

## -- Column specification -----
## Delimiter: ","
## chr (1): department
## dbl (4): employee_id, engagement, salary, vacation_days_taken
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

head(survey)

## # A tibble: 6 x 5
##   employee_id department engagement salary vacation_days_taken
##       <dbl> <chr>         <dbl>   <dbl>         <dbl>
## 1         1 Sales             3 103264.             7
## 2         2 Engineering        3  80709.            12
## 3         4 Engineering        3  60737.            12
## 4         5 Engineering        3  99116.             7
## 5         7 Engineering        3  51022.            18
## 6         8 Engineering        3  98400.             9
```

```
# Get an overview of the data
summary(survey)
```

```
##   employee_id      department      engagement      salary
##   Min.       : 1.0   Length:1470   Min.       :1.00   Min.       : 45530
##   1st Qu.: 491.2   Class :character   1st Qu.:3.00   1st Qu.: 59407
##   Median :1020.5   Mode  :character   Median :3.00   Median : 70481
##   Mean  :1024.9                      Mean  :3.05   Mean  : 74162
##   3rd Qu.:1555.8                      3rd Qu.:4.00   3rd Qu.: 84763
##   Max.    :2068.0                      Max.    :5.00   Max.    :164073
##   vacation_days_taken
##   Min.       : 0.00
##   1st Qu.: 6.00
##   Median :10.00
##   Mean  :11.27
##   3rd Qu.:16.00
##   Max.    :38.00
```

```
# Examine the counts of the department variable
survey %>% count(department)
```

```
## # A tibble: 3 x 2
##   department      n
##   <chr>      <int>
## 1 Engineering  961
## 2 Finance      63
## 3 Sales      446
```

```
# Output the average engagement score for each department, sorted
survey %>%
```

```
  group_by(department) %>%
  summarize(avg_engagement = mean(engagement)) %>%
  arrange(avg_engagement)
```

```
## # A tibble: 3 x 2
##   department avg_engagement
##   <chr>      <dbl>
## 1 Sales      2.81
## 2 Engineering 3.15
## 3 Finance      3.24
```

```
# Creating the disengaged variable
```

```
survey_disengaged <- survey %>%
  mutate(disengaged = ifelse(engagement <= 2, 1, 0))
```

```
survey_disengaged
```

```
## # A tibble: 1,470 x 6
##   employee_id department engagement salary vacation_days_taken disengaged
##   <dbl> <chr>      <dbl> <dbl>      <dbl>      <dbl>
## 1         1 1 Sales      3 103264.      7          0
```

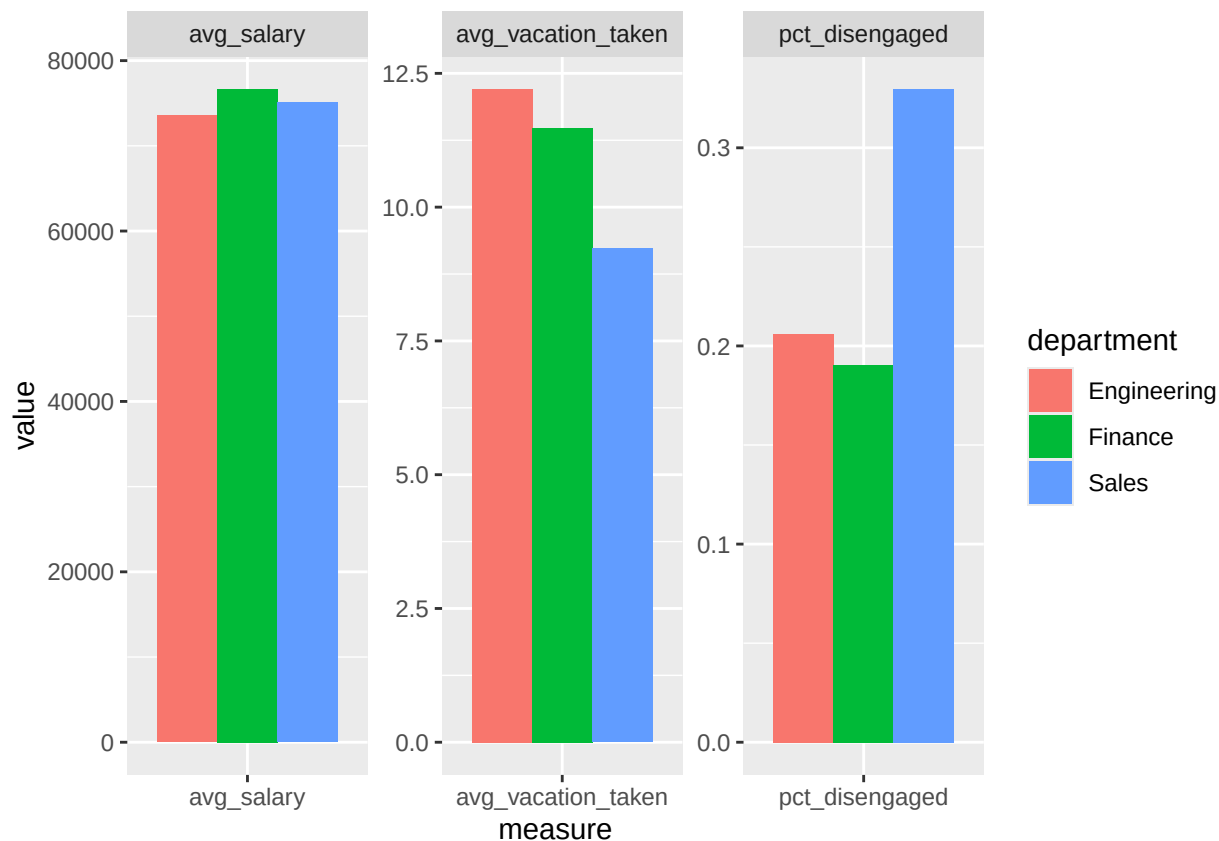
```
## 2      2 Engineering      3 80709.      12      0
## 3      4 Engineering      3 60737.      12      0
## 4      5 Engineering      3 99116.       7      0
## 5      7 Engineering      3 51022.      18      0
## 6      8 Engineering      3 98400.       9      0
## 7     10 Engineering      3 57106.      18      0
## 8     11 Engineering      1 55065.       4      1
## 9     12 Engineering      4 77158.      12      0
## 10    13 Engineering      2 48365.      14      1
## # i 1,460 more rows
```

```
# Summarize the three variables by department
survey_summary <- survey_disengaged %>%
  group_by(department) %>%
  summarize(pct_disengaged = mean(disengaged) ,
            avg_salary = mean(salary),
            avg_vacation_taken = mean(vacation_days_taken))
survey_summary
```

```
## # A tibble: 3 x 4
##   department pct_disengaged avg_salary avg_vacation_taken
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 Engineering 0.206    73576.    12.2
## 2 Finance    0.190    76652.    11.5
## 3 Sales      0.330    75074.     9.22
```

```
# Gather data for plotting
survey_gathered <- survey_summary %>%
  gather(key = "measure", value = "value",
         pct_disengaged, avg_salary, avg_vacation_taken)

# Create three bar charts
ggplot(survey_gathered, aes(x=measure, y = value, fill = department)) +
  geom_col(position="dodge") +
  facet_wrap(~ measure, scales = "free")
```



```
# Add the in_sales variable
survey_sales <- survey_disengaged %>%
  mutate(in_sales = ifelse(department == "Sales", "Sales", "Other"))

# Test the hypothesis using survey_sales
chisq.test(survey_sales$in_sales, survey_sales$disengaged)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: survey_sales$in_sales and survey_sales$disengaged
## X-squared = 25.524, df = 1, p-value = 4.368e-07
```

```
# Test the hypothesis using the survey_sales data
t.test(vacation_days_taken ~ in_sales, data = survey_sales)
```

```
##
## Welch Two Sample t-test
##
## data: vacation_days_taken by in_sales
## t = 8.1549, df = 1022.9, p-value = 1.016e-15
## alternative hypothesis: true difference in means between group Other and group Sales is not equal to
## 95 percent confidence interval:
## 2.229473 3.642409
## sample estimates:
```

```
## mean in group Other mean in group Sales
##      12.160156      9.224215
```